

Option D: $PV = nRT$. Same T thus same PV .

14 Ans: A

velocity is given by the $v = ds/dt$ thus the gradient of the displacement- time graph
acceleration is given by $a = -\omega^2 x$ thus a is always opposite to x

Option A: negative velocity and negative acceleration (v and a same direction)

Option B: negative velocity and zero acceleration (v and a opp direction)

Option C: negative velocity and positive acceleration (v and a opp direction)

Option D: zero velocity and positive acceleration (v and a opposite direction)

15 Ans: D